

Synthetic Latent Scaffolds: Sequential LatentMAS without Upstream Agents

Anonymous ACL submission

Abstract

Latent multi-agent systems such as LatentMAS replace text messages with a shared key-value (KV) cache: silent Planner, Critic, and Refiner agents grow a relay of latent steps, and only a Judger decodes. That upstream pass is extra compute—three agents $\times (K+1)$ latent forwards—before any tokens appear. We take sequential LatentMAS as given, including its published gains over a single agent, and ask whether those silent agents must still run *on the test item*.

We replace them with one frozen, question-independent KV prefix per problem type (**SynthScaffold**). The recipe is allowed to depend on the type. On multiple-choice MedQA we pool per-(layer, head, channel) magnitudes from a few training relays and sample an i.i.d. Gaussian tensor of typical length. On GSM8K we do not average finished KV tables: we average the K latent embeddings of each silent role, restore typical length, and replay that tape on a type prompt so the *model* writes the prefix.

On Qwen3-14B, greedy decode, $K=10$, generate_bs=20, Judger budget 1024: MedQA $n=40$ ties full LatentMAS at 67.5% (empty-cache 32.5%); GSM8K $n=40$ Mean-Replay ties at 90.0% (empty-cache 80.0%). Planner, Critic, and Refiner are never run at test time. We do not claim a KV-memory win, Qwen3-4B, or contest-math accuracy from a long-decode batched harness that under-scores LatentMAS itself.

1 Introduction

Multi-agent prompting lets models play specialized roles and exchange text (Du et al., 2023; Wu et al., 2023; Wei et al., 2022). Zou et al. (2025) push those messages *inside* the transformer: three upstream agents run a fixed number K of latent forward passes, emit no tokens, and append to a shared KV cache (past_key_values).

A final Judger attends to that cache and generates the answer. On Qwen3-14B their sequential system improves over a single-model baseline on GSM8K (95.2 vs. 83.7), MedQA (80.7 vs. 64.7), and AIME24 (66.7 vs. 63.3), while using fewer tokens than text-mediated MAS. We do not reopen that comparison. This paper is about skipping the silent-agent tax— $3(K+1)$ forwards, 33 at $K=10$ —while staying on the LatentMAS side of it.

That tax is worth paying on every test item only if the Judger needs those agents’ *instance-specific* thoughts. A diagnostic we put in Appendix B says otherwise on MedQA and GSM8K: replacing an item’s real cache with another item’s real cache tracks the original, while removing the cache hurts when the Judger is short of tokens. The relay behaves like a *type-level decoding prefix*—a shared object for “this kind of problem”—not a message about this x . That observation is motivation, not the method, and it is not a claim that latent collaboration is useless: a type-level prefix is still a latent prefix.

Method. **SynthScaffold** is one frozen KV prefix per problem type, built from training-set relays and reused for every test item. No Planner, Critic, or Refiner at inference. No training of the host. Different types may use different constructions (§3). MedQA uses an i.i.d. Gaussian tensor matched to per-channel KV magnitudes. GSM8K uses **Mean-Replay**: average donor latent embeddings *by step*, restore median L2, replay once on a type prompt so the model writes KV. Averaging finished KV tables is *not* the math recipe; it collapses GSM8K far below a Judger with no cache.

What this paper is. A methods paper: skip sequential LatentMAS’s upstream pass on a verified fork (Qwen3-14B; Yang et al., 2025) by substituting a type-level prefix. We report accuracy, Judger length, and wall-clock against full LatentMAS and

against an empty-cache control that uses the same Judger prompt and token cap.

What this paper does not claim. We do not skip Judger compute. We do not shrink KV memory: the prefix is full length. We do not beat full LatentMAS accuracy as a goal; on MedQA and GSM8K we match it at zero upstream forwards. We do not claim that LatentMAS fails to beat a single agent—that would contradict Zou et al. (2025) and our own Real > None gaps. We do not claim Qwen3-4B. We do not claim the appendix swap test as a contribution. We do not put AIME in the main table: a type-level prefix does not recover silent-agent accuracy there, even after Real is restored at batch size 1 (Appendix D).

Contributions.

1. **SynthScaffold:** a question-independent, training-free KV prefix that replaces Planner/Critic/Refiner at inference, with a type-dependent recipe (§3).
2. **MedQA:** an i.i.d. Gaussian prefix matches full LatentMAS at 0 silent-agent forwards (§4).
3. **GSM8K:** Mean-Replay matches full LatentMAS; the MedQA Gaussian recipe does not transfer; averaging finished KV tensors fails (§4).

2 Sequential LatentMAS

Pipeline. Sequential LatentMAS runs three silent agents then a text Judger. $\text{Planner}_K \rightarrow \text{Critic}_K \rightarrow \text{Refiner}_K$ grow a shared relay C , and only the Judger decodes. Each silent agent realigns its last-layer hidden state with W_a and runs K extra forwards, appending KV slots. At $K=10$ the system therefore spends 33 latent forwards per item before the Judger starts. A relay is not a short “thought vector”: each silent agent first prefills its role prompt (which includes the question) and then writes K latent steps, so the cache is mostly prompt tokens plus $3K$ latent positions.

Three systems we compare. Let C_x be the relay produced by Planner/Critic/Refiner on question x .

- **Real** (full LatentMAS): decode $p_\theta(y \mid x, C_x)$.
- **SynthScaffold** (ours): decode $p_\theta(y \mid x, \hat{C}_\tau)$, where $\hat{C}_\tau$ depends on the problem type τ but not on the test item x .

- **None:** decode $p_\theta(y \mid x)$ with an empty past, using the sequential Judger prompt and the same $T_{\max}$.

None is a *same-prompt empty-cache* ablation, not a re-run of the original paper’s baseline single-agent script. Zou et al. (2025) already show sequential LatentMAS above that Single baseline (their Tables 1–2; temperature 0.6, full evaluation sets). Our Real vs. None comparison is the same *direction* on MedQA and GSM8K. The absolute Single numbers are not interchangeable with None: different prompt family, greedy vs. sampled decode, and (in the main table) a matched 1024-token cap. HuggingFace generate always has a token ceiling (max_new_tokens; the fork default is 4096, README examples often use 2048). We did not invent that cap. We *match* it across the three arms and, in the main table, set it to $T_{\max}=1024$ so the comparison is about the prefix under a shared budget.

Fork. Code is a byte-for-byte copy of Gen-Verse/LatentMAS at commit 9a9e4d3 (24/24 file SHAs). At $K=40$, sequential HF decode, Qwen3-14B, a one-seed $n=150$ check sits within 1–2.5 points of their published GSM8K / ARC-Challenge / MedQA table means (Appendix A). On AIME24, their official sequential path at batch size 1 reproduces their 66.7% LatentMAS number (Appendix D). SynthScaffold is layered on that path: we do not reimplement the agents; we omit them.

Default operating point. Unless noted: Qwen3-14B, $K=10$ for the full-LatentMAS baseline, greedy Judger decode (temperature 0), $T_{\max}=1024$, generation batch size $B=20$ (the fork’s generate_bs). Tasks in the main table: MedQA (Jin et al., 2021) and GSM8K (Cobbe et al., 2021). Short-decode settings at this batch size are the claims. Long-decode contest math is discussed only as an evaluation caveat (Appendix D).

3 SynthScaffold

Full LatentMAS decodes $p_\theta(y \mid x, C_x)$. SynthScaffold decodes $p_\theta(y \mid x, \hat{C}_\tau)$ with a prefix built once per type τ and frozen. Offline, pick $N=8$ training questions of type τ , disjoint from the evaluation slice, and run sequential Planner $\rightarrow$ Critic $\rightarrow$ Refiner at the same K as the Real baseline.

MedQA: Gaussian magnitudes. Do not line donor caches up and average slot t of donor 1 with slot t of donor 2. For each (layer, key or value, head, channel), pool *every* time index of *every* donor into one scalar mean μ and standard deviation σ . The time axis is discarded: position 1 and position 400 contribute to the same bucket. Let L be the median donor length. Sample one tensor $\hat{C}_\tau$ of length L : every position is an independent draw

$$\hat{K}_t, \hat{V}_t \sim \mathcal{N}(\mu, \text{diag}(\sigma^2)), \quad t = 1, \dots, L. \quad (1)$$

The same (μ, σ) is used at every t . Freeze $\hat{C}_\tau$. At test time the Judger attends to that tensor, then to the test question’s Judger prompt. Planner, Critic, and Refiner are not run.

This object is *stat-matched noise of typical length*, not a thought about the test item and not an average of eight thoughts. Appendix C records tensor axes for Qwen3-14B.

Why Gaussian is enough on MedQA. MedQA is four-way multiple choice. Appendix B shows that a *wrong* item’s real cache helps about as much as the right one: the Judger is not reading medical content out of the relay. A long prefix with typical KV magnitudes is enough to put a 14B Judger into letter-mode (pick A/B/C/D, stop). i.i.d. Gaussian noise with those magnitudes is that prefix.

GSM8K: Mean-Replay. Open-ended arithmetic is a different decoding mode. Single-agent is already strong; the extra LatentMAS points are not “enter letter-mode.” The Gaussian object of the previous paragraph therefore need not transfer, and in Table 1 it does not.

Mean-Replay keeps a *sequence* of latent steps without averaging finished KV tables. From the same $N=8$ training donors, save the K last-layer latent embeddings that each silent role feeds as the next input ($K=10$ steps $\times$ 3 roles, hidden size 5120 on Qwen3-14B). Average those embeddings *by step* across donors, then rescale each step to the median donor L2. Replay the three mean tapes once on a type prompt that names the problem class but not a test item (“grade-school math word problem ... boxed numeric answer”). The model writes KV from that replay. Freeze the resulting cache. Averaging is in embedding space; the prefix the Judger sees is model-written KV, not a blend of eight KV tensors.

What not to do. Right-aligning finished donor caches and averaging over the donor axis destroys

Setting	Recipe	n	Real	Ours	None
MedQA $T=1024$, $B=20$	Gauss.	40	67.5	67.5	32.5
MedQA $T=1024$ (pilot)	Gauss.	80	60.0	53.7	33.8
GSM8K $T=1024$, $B=20$	Replay	40	90.0	90.0	80.0
GSM8K $T=1024$, $B=20$	Gauss.	40	90.0	82.5	80.0
GSM8K $T=1024$ (pilot)	Gauss.	60	93.3	88.3	80.0

Table 1: Accuracy (%). Real = sequential LatentMAS ($K=10$). Ours = frozen type-level prefix (0 upstream forwards). None = Judger prompt, empty past, same $T_{\max}$. Qwen3-14B, greedy. Gauss. = §3 Gaussian; Replay = Mean-Replay. The $B=20$ rows use the fork’s generate_bs; the $n=80/60$ rows are earlier $B=1$ pilots. MedQA $B=20$: mean Judger tokens 519 (Ours) vs. 704 (Real). GSM8K Mean-Replay: 612 (Ours) vs. 569 (Real) vs. 674 (None); end-to-end 2.84s vs. 3.12s vs. 2.14s per item. Real > None on both tasks, in the same direction as Zou et al. (2025).

the trajectory: on GSM8K $n=150$ that KV-mean scores 24.7% against Real 90.7% and None 82.0% (Table 2). A type-level prefix for math has to remain something the model could have written.

Type-dependent recipes. Nothing in the system statement requires every τ to use the same sampler. This draft specifies Gaussian for MedQA and Mean-Replay for GSM8K. Contest math (AIME) is a mixture of subjects, not one type; a subject-conditional construction is sketched in Appendix D and is not a main-table result.

4 Main results

Table 1 is the methods comparison: same Judger, matched budget, with or without the three silent agents. Ours uses the type’s specified recipe, not one object on every row.

MedQA. At the fork batch size, the Gaussian prefix *ties* full LatentMAS (67.5%) and more than doubles None (32.5%). Silent-agent wall-clock is already small at $B=20$ (≈ 0.17 s/item); the measured end-to-end win is a shorter Judger (4.26s vs. 5.40s) plus zero Planner/Critic/Refiner forwards. Precompute on 8 donors was 12.9s once. This is the result the Gaussian recipe is for: same accuracy, skip the upstream pass. The 67.5% operating point is a tight $T_{\max}=1024$; at 4096 tokens on the same backbone the Real arm is 80.0% (Appendix B), in line with their published sequential MedQA 80.7%.

GSM8K (Mean-Replay). Averaging latent embeddings by step and letting the model write KV ties full LatentMAS (90.0%) and sits 10 points

above None (80.0%). Paired Ours—Real correctness is 0 (bootstrap CI crosses 0). Precompute on 8 donors was 9.2s once. The same $n=40$ slice with the MedQA Gaussian object is 82.5%: above None, well below Real. We report that row as a failed transfer of the MedQA recipe, not as the math system.

Relation to the original tables. Zou et al. (2025) evaluate sampled decode (temperature 0.6, top- p 0.95) on full sets. Our main table is greedy, $n=40$, $T_{\max}=1024$. Those are different protocols; they are not a correction of their numbers. What we need for the method is that *on this protocol* Real still beats None and Ours matches Real. Appendix A is the check that the Real arm is still their system.

Robustness. Three independent Gaussian seeds on MedQA $n=40$, $T_{\max}=1024$: 55/60/65, all well above None (≈ 35). The MedQA tie at $B=20$ is not one lucky draw.

Compute. Real: 33 latent forwards per item plus Judger decode. SynthScaffold: 0 latent forwards per item; the prefix is built once offline. KV length is unchanged, so the win is *upstream compute*, not batched KV memory.

5 What has to stay in the prefix

Table 2 asks which objects reproduce the 14B MedQA effect at $T_{\max}=1024$, $n=40$ (an earlier diagnostic slice: Real 67.5, None 35.0), and records the GSM8K constructions that are not Mean-Replay.

Length and rank. Short fixed synths of length 32 or 64 still recover most of the MedQA benefit on this slice (≈ 57.5); rank 32–64 at full length is in the same band. Repeat-last and trunc-64 of a *real* cache fail. The deployed method therefore keeps full length: we save the 33 forwards, not the KV footprint.

Learned short slots. Distilling $m=64$ learnable KV slots to full-LatentMAS Judger behavior reaches 50%—above None, below the frozen full-length synth. The training path works (loss 0.08, gradients reach the slots). Compressing the scaffold to a handful of slots is not yet a better method than sampling one long statistical prefix.

Qwen3-4B. On 4B MedQA $n=40$, $T_{\max}=1024$, only real and shuffled caches work (45%).

Variant	Acc.	Up. fwds
Real cache	67.5	33
Shuffled real cache	62.5	33
Per-item matched noise	60.0	33
PCA rank-32 of real	65.0	33
Fixed Gaussian (MedQA)	60.0	0
Repeat-last (one vector)	22.5	33
Truncate to 64 slots	40.0	33
Learned $m=64$ slots	50.0	0
None	35.0	0
GSM8K Mean-Replay $n=40$	90.0	0
GSM8K Gaussian $n=40$	82.5	0
GSM8K KV-mean $n=150$	24.7	0
GSM8K Real/None $n=150$	90.7/82.0	33/0

Table 2: Upper block: 14B MedQA, $T_{\max}=1024$, $n=40$ (diagnostic slice). Coarse full-length statistics work; a single vector and a short truncation do not. The 60.0 fixed-Gaussian row is this older slice; Table 1 is the $B=20$ operating point (67.5). Lower block: GSM8K recipes under the same $T_{\max}=1024$ greedy protocol. Mean-Replay ties Real; averaging finished KV tables does not.

Matched noise, PCA, and truncation collapse toward None or worse. SynthScaffold with the Gaussian construction is a 14B method.

6 Related work

Latent multi-agent systems. LatentMAS (Zou et al., 2025) is the system whose upstream we replace. We inherit their sequential pipeline, their finding that latent collaboration beats a single agent and text MAS, and their HuggingFace KV-handoff implementation. Coconut (Hao et al., 2024) trains continuous latent thoughts inside one model; we leave the host frozen and substitute a synthetic KV prefix for three silent roles. Text MAS (Du et al., 2023; Wu et al., 2023) exchanges language; we ask whether the latent stand-in for those messages can be a single shared scaffold.

KV statistics, not KV eviction. StreamingLLM attention sinks (Xiao et al., 2023) and H2O heavy hitters (Zhang et al., 2023) compress a cache that was already built for this sequence. SynthScaffold never builds that cache at test time. The related fact we use is weaker: a long prefix with the right per-channel moments (MedQA), or a model-written replay of mean latent steps (GSM8K), is enough to induce the Judger’s type-level mode.

Steering. Activation and KV steering (Turner et al., 2023; Rimsky et al., 2024; Chen et al., 2025; Li et al., 2025) act at the text-emitting Judger.

They are complementary: they do not create the prefix, and they do not replace the silent agents.

7 Conclusion

SynthScaffold is a training-free substitute for sequential LatentMAS’s silent agents: one frozen, type-level KV prefix, with a recipe that is allowed to depend on the problem type. On Qwen3-14B MedQA, an i.i.d. Gaussian prefix with typical KV magnitudes matches full LatentMAS accuracy at **zero** upstream compute. On GSM8K, Mean-Replay—average latent embeddings by step, let the model write KV—does the same. The MedQA Gaussian object does not transfer to GSM8K, and averaging finished KV tensors fails. That is evidence for type-dependent prefixes, not evidence against LatentMAS. The practical lever is to skip the per-item relay after a one-time precompute—not to treat Planner/Critic/Refiner as a necessary per-item message channel, and not to claim the original collaboration gains were illusory.

Limitations

One backbone where the specified constructions work (Qwen3-14B); Qwen3-4B fails with the Gaussian object. Greedy decode, $K=10$ Real baseline, $n=40$ in the main table. The prefix is full length, so Judger attention cost is not reduced. Learned short slots do not yet match the fixed prefix. Hierarchical LatentMAS is not evaluated. Results replace the upstream of one published sequential system; they do not revise that system’s comparison to Single or TextMAS. AIME24 (Maxwell-Jia, 2024) is not in the main table. At batch size 1 the silent agents recover Gate-1 / Zou 66.7%; a MATH Level-5 subject-medoid prefix does not (56.7% vs. None 53.3%; Appendix D). That is a failed transfer, not a harness artifact.

Ethics statement

This work replaces the silent agents of a public multi-agent inference method with a synthetic KV prefix and evaluates on standard academic benchmarks. It does not release new models or user data. The main risk is deploying the Gaussian scaffold on small models, as a universal cache, or from a long-decode batched harness that does not reproduce LatentMAS; Table 1, the 4B collapse, and Appendix D already say those are not licensed.

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A Fork check

Task	Single*	LatentMAS*	Fork
GSM8K	83.7	95.2	92.7
ARC-Challenge	92.6	95.6	94.7
MedQA	64.7	80.7	79.3
AIME24	63.3	66.7	66.7 [†]

Table 3: Sequential Qwen3-14B. *Zou et al. (2025) Tables 1–2 (sampled decode, full sets, 3-seed means). Fork GSM8K / ARC-C / MedQA: $K=40$, $n=150$, 1 seed, sequential HF. [†]AIME24: official LatentMASMethod, $K=10$, greedy, $T_{\max}=8192$, $B=1$, $n=30$ (artifacts/gate1/aime2024_tok8192). Gaps on the $n=150$ slice are consistent with subset and seed, not a different algorithm. We inherit LatentMAS > Single; we do not replace that table.

B Why a shared cache is allowed

The method in the main paper is a type-level prefix. This appendix is the justification that instance-specific content is not what the Judger uses—not a contribution of its own.

Decode the Judger from $C_{x'}$ for a different same-task x' , or from no cache. If C carried x -specific intermediate reasoning, accuracy should track whether $x' = x$.

Table 4 reports that test on Qwen3-14B, greedy, $n=40$ MedQA. At $T_{\max}=1024$, removing the cache drops accuracy from 67.5% to 35.0%: *presence* of a latent prefix matters under a tight budget. Replacing it with another item’s cache stays close to the real cache (62.5% vs. 67.5%). At 2048 and 4096 tokens the Judger can re-solve; real and shuffled are tied, and the gap to no-cache shrinks. That

crossover is a measurement on a cap LatentMAS already uses (max_new_tokens), not a cap we introduced. At 4096, Real 80.0% matches their published sequential MedQA mean (80.7%) on this small slice; None 75.0% is still a same-prompt empty-cache arm, not their Single baseline.

GSM8K at $T_{\max}=1024$, $n=60$ has the same content-independence signature (real 93.3%, shuffled $\approx 93\%$, none 80%), with mean Judger tokens 542/560/668: here None is not starved, so the prefix still adds a content-independent accuracy bump.

Linear and MLP probes of the gold answer from per-agent hidden states are near chance (four-way MedQA, ≈ 0.28 – 0.33 vs. 0.25). Attempts to steer the latent channel for accuracy (diff-of-means residuals, KV-cache steering of handoff columns, a learned rank objective) do not beat an unsteered baseline; Judger-only concision steering *does* change length (Chen et al., 2025; Li et al., 2025). Those negatives are why we *replace* the tape with a type-level object rather than try to enrich it.

$T_{\max}$	Real	Shuffled	None	Tok. (R/S/N)
1024	67.5	62.5	35.0	722 / 712 / 916
2048	77.5	77.5	70.0	882 / 867 / 1329
4096	80.0	77.5	75.0	892 / 878 / 1603

Table 4: 14B MedQA, $n=40$, greedy. Shuffled = another item’s real cache (rolling derangement). A wrong-item cache tracks the real cache; no cache hurts mainly when the Judger is short of tokens. Motivation for a shared prefix, not the method.

C Gaussian and Mean-Replay details

Gaussian (MedQA). Donors: $N=8$ training-split sequential LatentMAS caches. Qwen3-14B stores 40 layers, 8 KV heads, head dimension 128. For each layer and for keys and values separately, compute mean and std over the position axis *and* across donors, keeping the (head, channel) axes. Sample $\mathcal{N}(\mu, \sigma^2)$ i.i.d. over positions at median donor length L . This is the object behind Table 1’s MedQA 67.5% tie.

Mean-Replay (GSM8K). Same $N=8$ train donors, $K=10$. Capture the latent embeddings of shape $[K, D]$ actually fed at each silent step ($D=5120$), not the finished KV table. Average over donors at each step; rescale to the median donor L2. Replay the three role tapes on the

type prompt, once, with the host frozen except for the ordinary latent-forward write. The frozen prefix in the GSM8K 90.0% row had length 446 (≈ 70 MB). Do not substitute a right-aligned mean of the eight KV tables; that is the 24.7% row in Table 2.

D Evaluation protocol: short decode vs. AIME

The original sequential protocol uses temperature 0.6, top- p 0.95, and (for AIME24/25) a 20,000-token cap (Zou et al., 2025). Our main table uses greedy decode and $T_{\max}=1024$ because that is where a type-level prefix is a decision, not a rounding error, and because MedQA/GSM8K answers fit. Those choices are stated; they are not a claim that sampled full-set LatentMAS numbers were wrong.

Where the latency harness is trusted. On MedQA and GSM8K at $B=20$, $T_{\max}=1024$, Real beats None by a large margin and matches the direction of their LatentMAS $>$ Single tables. Judger lengths are hundreds of tokens, not thousands. That is the operating point behind Table 1.

Where it is not. AIME24 is 30 contest problems (Maxwell-Jia, 2024). Their published sequential 14B numbers are LatentMAS 66.7 vs. Single 63.3—a small absolute gap at high accuracy, not a collapse of either arm. Our official-path Gate-1 run (LatentMASMethod, $B=1$, greedy, $T_{\max}=8192$, $n=30$) is $K=10$ 66.7% vs. $K=0$ 53.3%. $K=0$ still prefills Planner/Critic/Refiner (3 latent forwards); it is not None. That Gate-1 $K=10$ number matches their LatentMAS AIME24 cell.

The same 30 items in scripts/exp_synth_latency.py at $B=8$, $T_{\max}=8192$, dropped Real to 33.3% and None to 33.3% (medoid Ours 30.0%). Mean Judger tokens were still $\sim 6k-7k$: long, wrong solutions. That was a left-padded long-decode batch, not evidence that LatentMAS equals a single agent on AIME.

Trusted $B=1$ re-run. The same script at $B=1$ recovers Gate-1 generation exactly (30/30 predictions and token counts). Integer grading (025=25) gives Real 20/30 = 66.7%. None is 16/30 = 53.3% (the string grader had been 13/30 and 10/30). Subject-conditional MATH Level-5 medoid replay (Hendrycks et al., 2021) — 8

donor tapes per subject, keep the medoid, replay on a contest type prompt, assign test items by last-token hidden cosine — scores 17/30 = 56.7%. Real uniquely solves items 4, 10, and 18; Ours never uniquely beats Real. AIME is not one problem type, and this prefix does not replace the silent agents. We leave it out of Table 1 as a failed transfer, not as an unevaluated construction.